

Chapter C4: Material choices

Overview

Our society uses a large range of materials and products that have been developed, tested and modified by the work of chemists. Materials used to make a particular product need to meet a specification which describes the properties the material needs to make it suitable for a particular use. This chapter looks at a range of different materials and investigates their properties in the context of their suitability for making consumer products. The chapter also considers how the life cycle of a product is assessed in its journey from raw material to final disposal.

Topic C4.1 considers the variety of materials that we use. Learners use data and information about the properties of 'pure' and composite materials to consider their suitability for making consumer products. Ceramics, glass, materials with giant structure and polymers are all considered.

Topic C4.2 moves on to look in detail at materials made using polymers. The topic covers the chemical reactions that happen when addition and condensation polymers form, and also studies naturally occurring polymers. This topic links

closely to the discussion of bonding and structure in Topic C4.3.

Topic C4.3 extends the study of properties to looking at bonding and structure in order to explain why a particular material behaves as it does. Learners learn about the bonding in metals, polymers and giant covalent structures and link the bonding and structure to the properties of the materials. They consider the usefulness of diagrams and models of bonding and structure to chemists who need to investigate and predict properties of materials so that they can make judgements about their usefulness or model likely changes in their properties if their structures are modified. A range of materials are studied, including new materials such as fullerenes and graphene.

Topic C4.4 looks specifically at the nature and uses of nanoparticles.

Topic C4.5 considers the life cycle of materials. The learners learn about the corrosion of metals such as iron and look at ways of extending the life of metal products by working to prevent corrosion. They also learn how the impact of our manufacture, use and disposal of consumer products is assessed using life cycle assessments.

Learning about material choices before GCSE (9–1)

From study at Key Stages 1 to 3 learners should:

- distinguish between an object and the material from which it is made
- identify and name a variety of everyday materials, including wood, plastic, glass, metal, water, and rock
- describe the simple physical properties of a variety of everyday materials
- compare and group together a variety of everyday materials on the basis of their simple physical properties.
- have observed that some materials change state when they are heated or cooled, and measured the temperature at which this happens in degrees Celsius (°C)
- compare and group together everyday materials on the basis of their properties, including their hardness, solubility, transparency, conductivity (electrical and thermal), and response to magnets
- identify and compare the suitability of a variety of everyday materials, including wood, metal, plastic, glass, brick, rock, paper and cardboard for particular use
- know the differences between atoms, elements and compounds
- recognise chemical symbols and formulae for some elements and compounds
- know about the properties of ceramics, polymers and composites (qualitative).

Tiering

Statements shown in **bold** type will only be tested in the Higher Tier papers.

All other statements will be assessed in both Foundation and Higher Tier papers.

Learning about material choices at GCSE (9–1)

C4.1 How is data used to choose a material for a particular use?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>
Our society uses a large range of materials and products developed by chemists. Chemists assess materials by measuring their physical properties, and use data to compare different materials and to match materials to the specification of a useful product (1aS4). Composites have a very broad range of uses as they allow the properties of several materials to be combined. Composites may have materials combined on a bulk scale (for example using steel to reinforce concrete) or have nanoparticles incorporated in a material or embedded in a matrix. The range of uses of metals has been extended by the development of alloys. Alloys have different properties to pure metals due to the disruption of the metal lattice by atoms of different sizes. Chemists can match an alloy to the specification of properties for a new product.	1. compare quantitatively the physical properties of glass and clay ceramics, polymers, composites and metals, including melting point, softening temperature (for polymers), electrical conductivity, strength (in tension or compression), stiffness, flexibility, brittleness, hardness, density, ease of reshaping
	2. explain how the properties of materials are related to their uses and select appropriate materials given details of the usage required
	3. describe the composition of some important alloys in relation to their properties and uses, including steel (<i>separate science only</i>)

Linked learning opportunities

Practical work:

- Practical investigation of a range of materials leading to classification into categories.

Ideas about Science:

- The range of materials developed by chemists enhances the quality of life. (1aS4)
- Use and limitations of a model to represent alloy structure. (1aS3)